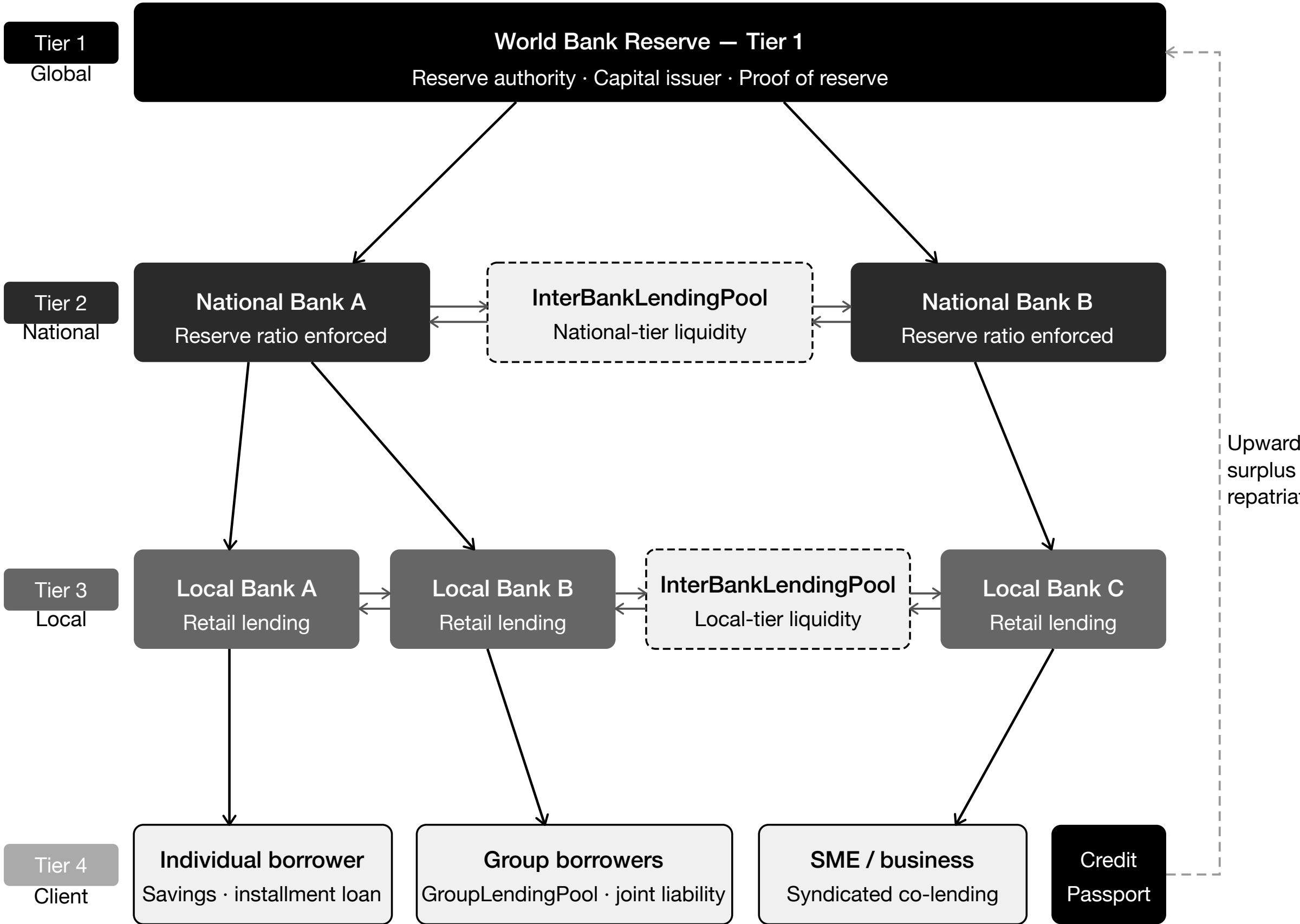




Six banking functions enforced on-chain:

- Deposit mobilization
- Credit allocation
- Payment & settlement
- Risk intermediation
- Liquidity management
- Ancillary services

Technology stack:

- L4

React 18 / TypeScript UI · wallet signing · optional MCP agent (Qwen3-8B, 17 banking tools)
- L3

Express.js + FastAPI · scikit-learn fraud detection · SHAP explainability · Chainlink Functions oracle
- L2

Solidity 0.8.20 smart contracts · ERC-4337 Paymaster · ZKP identity (Circom 2.0 / Groth16) · RBAC
- L1

Polygon zkEVM Cardona · Ethereum Sepolia · PostgreSQL (19 entities) · IPFS

Flow key:

- allocateCapital — downward capital allocation (Tier 1 → 2 → 3 → 4)
- ↔ InterBankLendingPool — same-tier liquidity (dashed border)
- - - -> UpwardDepositFacility — surplus repatriation to parent tier